

Java Enterprise Technologies

Question Bank | 2 Sets × 10 Questions

Networking • JDBC • Servlets • JSP • JSF • Hibernate • Spring MVC

SET 1 — Question Paper

Instructions: Answer all 10 questions. Each question carries equal marks. Support answers with code/diagrams where needed.

Q.No	Topic	Question
1	Java Networking	What is a Socket in Java? Write a simple Java program to create a TCP/IP client socket that connects to a server and sends a "Hello" message.
2	JDBC	What is JDBC? List the steps to connect to a database using JDBC and write a simple Java program to connect to a MySQL database and execute a SELECT query.
3	Servlet API	Explain the three phases of the Servlet life cycle: init(), service(), and destroy(). What is the role of web.xml (deployment descriptor) in a web application?
4	Java Networking	What is a Datagram in Java? Differentiate between TCP and UDP. Write a short Java program using DatagramSocket to send a message from client to server.
5	JSP	What are the problems with Servlets that JSP solves? List and briefly explain any three JSP Implicit Objects (e.g., request, response, session) with examples.
6	Servlet API	What is a Filter in Servlets? Explain the methods of the Filter interface: init(), doFilter(), and destroy(). Give one example use case such as logging or authentication.
7	JSF 2.0	List the six phases of the JSF request processing life cycle. What is the purpose of JSF Validation Tags? Give one example of a validation tag used in a JSF form.
8	Hibernate	What is Hibernate and why is it used? Draw or describe the basic Hibernate architecture showing the key components: Configuration, SessionFactory, Session, and Transaction.
9	Spring MVC	What is Spring MVC? Describe the request flow in Spring MVC from the browser to the controller and back. What is the role of the DispatcherServlet?
10	Session Management	What is session tracking? Compare three session tracking techniques: Cookies, URL Rewriting, and HttpSession. When would you use each one?

SET 2 — Question Paper

Instructions: Answer all 10 questions. Each question carries equal marks. Support answers with code/diagrams where needed.

Q.No	Topic	Question
1	Java Networking	What is a ServerSocket in Java? Write a simple Java program to create a TCP/IP server that waits for a client connection and prints the received message. Also explain what InetAddress is.
2	JDBC	What is a ResultSet in JDBC? Explain the difference between Statement, PreparedStatement, and CallableStatement. Give a short code example using PreparedStatement to insert a record.
3	Servlet API	What is the RequestDispatcher interface? Explain the difference between forward() and include() with a simple example. How do you set and get an attribute in a Servlet request?
4	JDBC	What is the difference between SQLException and SQLWarning in JDBC? How do you handle SQL errors in Java? Write a short try-catch block showing proper JDBC error handling.
5	JSP	What is JSTL? List the commonly used JSTL core tags (e.g., <c:if>, <c:forEach>) with a short example. How do you handle exceptions in JSP using error pages?
6	JSP	How does JSP handle form data? Write a simple JSP example that reads form input (name and email) and displays it. Also explain how to create and read a Cookie in JSP.
7	JSF 2.0	What are JSF Facelets? List any four commonly used JSF standard UI components (e.g., h:inputText, h:commandButton). What is PrimeFaces and how does it extend JSF?
8	Hibernate	What is HQL (Hibernate Query Language)? How is it different from SQL? Write a simple HQL query to fetch all records from an Employee entity. What is a Hibernate Annotation?
9	Spring MVC	What is Dependency Injection in Spring? Explain with a simple XML bean configuration example. What is the difference between BeanFactory and ApplicationContext?
10	HTTP & Servlets	What are the HTTP methods GET and POST? How do doGet() and doPost() methods work in a Servlet? Write a simple Servlet that handles a GET request and returns an HTML response.